INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 3: Research Methodology

by

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3 Research Methodology

Chapter 2 ended with five gaps. This chapter gives each of them a measurable procedure and sets out how the coupled walk C-PR, the method of the thesis, and its two single-layer references, EndorseRank and Adaptive Weighted PageRank (AWP), are evaluated as reputation scores for contract accounts on Arbitrum One. EndorseRank runs on the ERC-20 allowance graph and AWP on the transfer graph (T. Do et al., 2023); C-PR walks both in one chain (Section 3.6.3). The two single-layer walks weight their edges as AWP's authors specify, and every walk is solved with weighted PageRank (Page et al., 1999); AWP restarts the walk in proportion to sending activity, and EndorseRank and C-PR restart it uniformly.

Methodologically, the study follows construct (domain) alignment. Reputation scores are continuous rankings, and the checks rest on observable contract-level statistics instead of binary default labels; each ranking method is one indicator of a reputation construct that cannot be observed directly, in the sense of construct validity (Cronbach & Meehl, 1955). The claim is comparative, every score set against the checks of each construct and against the raw degrees it smooths; it is not a prediction of trading profit or credit default.

Four principles guide each operational choice.

- Same-seed comparability: every score is computed on an identical set of contracts, so no difference between scores can come from sampling.
- Fixing choices in advance: the analysis plan and the versioned configuration were committed to the authors' Git repository at 13:41 UTC on 9 October 2026 (commit `d060f3a` of its public copy; Section 3.13), before the study extracted any log.[1] They fix the windows, the cohort and account-type rules, every score with its parameters, the values of λ, the proxies, the W0 labels, the bootstrap, the benchmark protocol, the robustness checks and rule A, the plan's one

[1] All times are UTC; the commits record UTC+9.

decision rule, whose object is C-PR (RQ4; Section 3.7). Scan A, the cohort, the ego extraction and the same-window and W0 results followed. At 16:49 UTC the same day the registration of W1 was committed (commit `1747e64`): rule B with its margin, F4–F6, two sensitivity analyses and the comparison on outcomes built from neither edge type, with the wording of its conclusions (Sections 3.7.2 and 3.7.3). Scan B, the registered evaluation, the supplementary contrasts, the robustness checks, the Sybil model and the benchmark came next. After these results were known, the revised analysis was defined (Section 3.5.1): amounts in US dollars, five tokens, and scores computed from the twelve months before each freeze. The revised analysis, which Chapter 4 reports, applies the same rules to these scores; every other contrast, the degree contrasts among them, was computed after its data were known and is reported as such, with its interval.

- Construction and timing: each cohort's edge lists are built once, before any timer starts, and the benchmarks time the solver call.
- Scope: H1 and the divergence half of C4 compare the orderings of EndorseRank and AWP; the other hypotheses and Claims C3, C5 and the agreement half of C4 concern every score, C1 and C2 the coupled walk and its ablation, and C6 the approval and transfer counts and their PageRanks on trader outcomes. RQ3–RQ5 draw on the spender cohorts of W0 and W1, and rule B adds a trader cohort.

3.1 Notation

The notation used in this chapter and in Chapters 4 and 5 is collected in Table 3.1 below, and a condensed list appears in Table II. Vectors are set in bold and sets in calligraphic type.

Table 3.1: Notation for graphs, scores, proxies, and parameters.

Symbol	Meaning
$\mathcal{W}$	Matched contract cohort ($\lvert\mathcal{W}\rvert = n = 6{,}483$)
$G = (V, E)$	Graph passed to the PageRank solver: V holds the end points of the retained edges E

Table 3.1: (continued)

Symbol	Meaning
$G_{\text{approve}} = (V_A, E_A, w_A)$	EndorseRank allowance graph
$G_{\text{transfer}} = (V_T, E_T, w_T)$	AWP transfer graph
$a^\star(u, v, \text{token})$	Latest in-window allowance of owner u to spender v on one token (US dollars)
$\Delta t^\star(u, v, \text{token})$	Age of the approval that set $a^\star$
$w_A(u, v)$	EndorseRank edge weight, owner $u \to$ spender v: $\sigma(\Delta t^\star)V(a^\star)$ summed over tokens
$w_T(u, v)$	AWP edge weight, sender $u \to$ receiver v: $\sigma(\Delta t_e)V(x_e)$ summed over transfers
x_e	Amount of transfer e (US dollars)
$V(z)$, b	Bounded value transform of AWP, $V(z) = 2/(1+e^{-bz}) - 1$, with $b = \ln 3/m$ (b_T for transfers, b_A for allowances)
m_T, m_A	Median transfer (170) and allowance (100) amounts in US dollars in the W0 score window
U	Cap on one allowance in C-PR and in the in-approve value (19,000 dollars)
X_u, X_u^A	Activeness of address u in the transfer and the allowance graph, the restart weights of AWP and of EndorseRank with activity restarts
$\tilde{w}_A(u, v)$, $\tilde{w}_T(u, v)$	Weights of the two layers of C-PR, in US dollars
$o(u)$; $o_A(u)$, $o_T(u)$	Out-strength $\sum_v w(u, v)$ of node u; the same in the allowance and transfer layers
P	Transition matrix, $P_{u,v} = w(u, v)/o(u)$; rows of dangling nodes are zero
$\mathbf{1}[\cdot]$	Indicator: 1 if the condition holds, 0 otherwise
$\mathbf{1}_{\text{dang}}$	Indicator vector of the dangling nodes, entries $\mathbf{1}[o(u) = 0]$
$\mathbf{r}$, $\mathbf{r}^{(t)}$	PageRank vector and its iterate after t steps
$\boldsymbol{\pi}$	Restart (teleportation) distribution, also used for dangling mass: uniform for EndorseRank and C-PR, proportional to X_u for AWP and to X_u^A for EndorseRank with activity restarts, and for S-PR the allowance layer's score restricted to V_T and renormalized
d	PageRank damping factor (default 0.85)
ε	PageRank convergence tolerance (10^{-8})
$I_{\max}$	Maximum PageRank iterations (300)
λ	Allowance-layer weight of the coupled operator C-PR ($\lambda = 0.5$ primary; 0.25, 0.75 sensitivity; at $\lambda = 1$ and $\lambda = 0$ one layer is walked alone)
$\alpha_A(u)$, $\alpha_T(u)$	Layer weights in node u's C-PR row; $D_\lambda(u)$ is their common denominator
$P^{(\lambda)}$	Coupled transition matrix of C-PR
$\mathbf{s}_M$	Reputation score vector for method M
$\mathbf{p}_q$	Proxy value vector for proxy q
$\sigma(\Delta t)$	Logistic time-decay on the age of an approval or transfer
k, t_0	Decay steepness (0.01) and midpoint (180 days)
Δt	Age of an event in days before the decay anchor (T_{obs}; t_1 in W0)
T_{obs}	Observation end timestamp and W1 freeze (2026-06-30 23:59:59 UTC)

Table 3.1: (continued)

Symbol	Meaning
t_1, t_2	W0 freeze (2026-03-31 23:59:59 UTC) and end of its label window (2026-06-30 23:59:59 UTC)
ρ	Spearman's rank correlation
R_i, S_i	Mid-ranks of contract i's score and proxy value (tied values share their average position)
τ, τ_b	Kendall's rank correlation; every reported value is the tie-corrected τ_b
n_c, n_d	Numbers of concordant and discordant contract pairs
n_0; n_1, n_2	All pairs, $n(n-1)/2$; pairs tied in the first and in the second ranking

Except where stated otherwise, PageRank means the weighted variant of Page et al. (1999), whose update at iteration t is

$$\mathbf{r}^{(t+1)} = (1-d)\,\boldsymbol{\pi} + d\,P^{\top}\mathbf{r}^{(t)} + d\left(\mathbf{r}^{(t)\top}\mathbf{1}_{\text{dang}}\right)\boldsymbol{\pi}, \tag{3.1}$$

where P is the transition matrix of Equation (3.2) below, $\boldsymbol{\pi}$ is the restart distribution over the nodes, uniform ($\pi_u = 1/|V|$) unless a method specifies another (AWP in Section 3.6.1, EndorseRank with activity restarts in Section 3.6.2, S-PR in Section 3.6.3), and $\mathbf{1}_{\text{dang}}$ is the indicator vector of the dangling nodes, with entries $\mathbf{1}[o(u) = 0]$. The third term therefore collects the mass held by nodes without out-edges. With $w(u,v)$ denoting the weight of the directed edge from u to v and $o(u) = \sum_v w(u,v)$ the out-strength, the transition matrix is

$$P_{u,v} = \begin{cases} w(u,v)/o(u) & \text{if } o(u) > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.2}$$

The rows of dangling nodes are zero, so P is substochastic; the mass lost at those nodes returns through the third term of Equation (3.1) and is spread according to $\boldsymbol{\pi}$. Filling each zero row with $1/|V|$ gives the row-stochastic matrix of Equation (2.1), and with uniform $\boldsymbol{\pi}$ Equation (3.1) is then the Google-matrix form of Equation (2.2). The iteration starts from $\mathbf{r}^{(0)} = \mathbf{1}/|V|$ and stops once $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon$ or after $I_{\max}$ iterations. V is the set of end points of the retained edges, and

the final vector is normalized to sum to one over V. Contracts are also listed in descending order of score, ties broken by address, but these ordinal ranks serve only for listing; the correlations of Section 3.10 are computed on the raw scores, where tied scores stay tied.

The damping factor d interpolates between pure propagation ($d \to 1$) and uniform scores ($d \to 0$). Its default of 0.85 is the classical web setting (Page et al., 1999), and the robustness sweep varies it with edge weights held fixed (Section 3.12). The strict $\varepsilon = 10^{-8}$ and $I_{\max} = 300$ keep the measured runtimes tied to the structure of the graph and not to early termination.

3.2 Data sources and sampling

This subsection fixes the chain, time interval, public tables, events and contracts of the main sample; no later step brings in events from outside it. Figure 3.1 places every kind of on-chain record used in the thesis on one calendar, with the months each analysis scores, the months it holds out as labels, and what each record is used for.

- Primary chain: All events come from Arbitrum One (chain ID 42161), an optimistic rollup on Ethereum (Thibault et al., 2022) whose logs are in the public BigQuery export. One chain avoids cross-chain aliases, where one user's addresses on several chains would appear as distinct nodes, and every address in the sample faces the same fees and latencies.
- Extraction and score windows: Scan A extracts the events from 2023-10-01 00:00:00 UTC to 2026-06-30 23:59:59 UTC (the queries take 2026-07-01 as an exclusive end), and scan B those of the label window W1, July to September 2026, so the data cover three years. Every score of the main analysis is computed from the twelve months before its freeze: from 2025-07-01 00:00:00 UTC to T_{obs} = 2026-06-30 23:59:59 UTC in the same-window evaluation and in W1, and from 2025-04-01 00:00:00 UTC to t_1 = 2026-03-31 23:59:59 UTC in W0 (Section 3.7). A window of one year bounds the age of every event that enters a score, a proxy, a baseline or a label history, gives the two freezes windows of the same length, and lets the tenure and active-month proxies range up to 365 days and 12 months. Both graphs

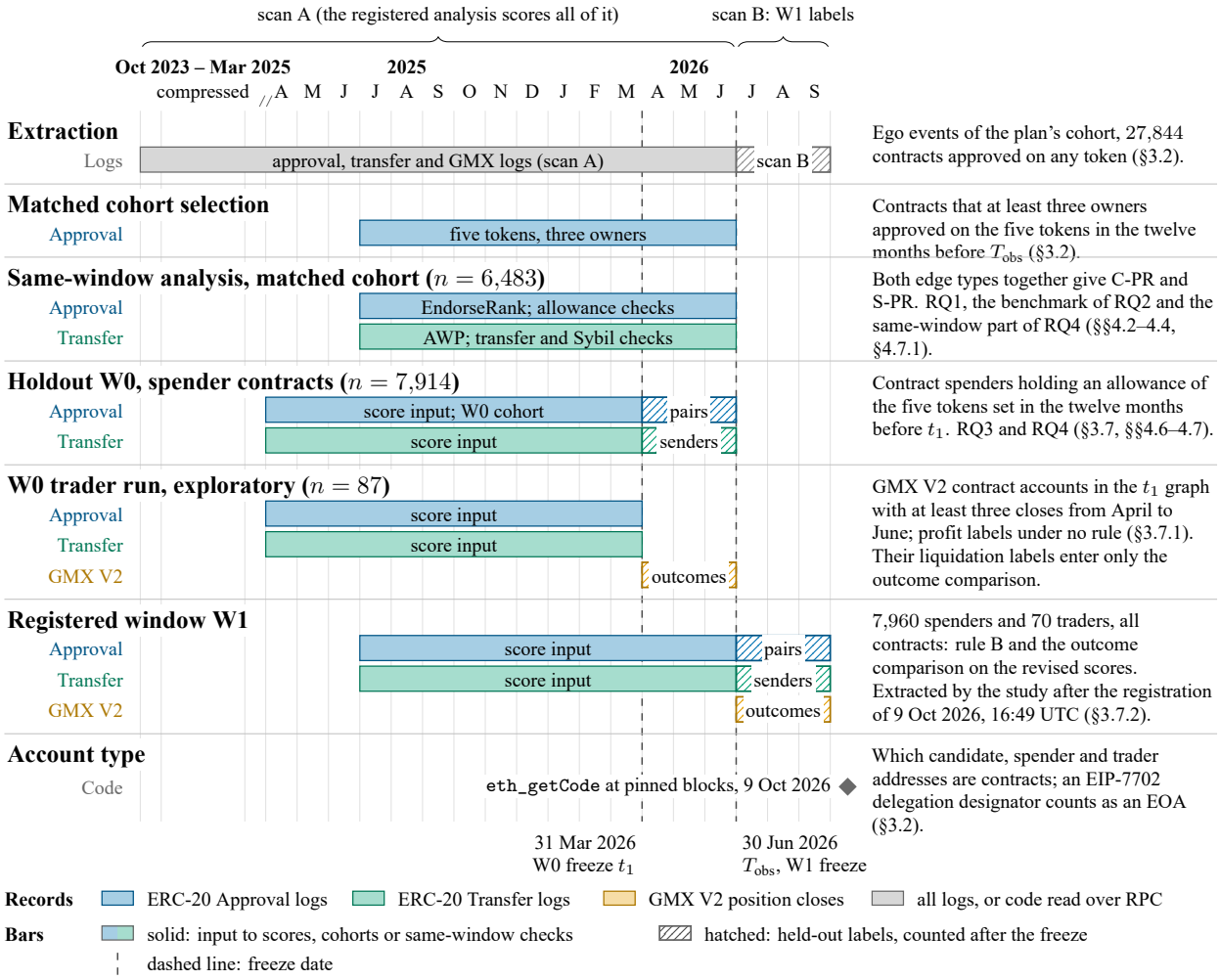


Figure 3.1: Blockchain data used in this thesis, all from Arbitrum One. Each lane is one analysis, and each bar is one kind of on-chain record placed on the months it covers; the 18 months before April 2025 are drawn at a compressed scale. Solid bars are inputs to scores, to cohorts or to same-window checks. Hatched bars are labels counted after a freeze date and kept out of the scores; the study extracted the July–September logs only after the registration of W1 was committed at 16:49 UTC on 9 October 2026. The text in a bar says what the records compute, and the right-hand column names the cohort, the research questions and the sections that report the results. The approval logs of five tokens in the twelve months before each freeze select the cohort of that freeze, and every approval and transfer query of the main analysis is then filtered to it, to those tokens and to those months; the registered analysis scores every event of scan A.

use exactly the same dates, and the time decay is anchored at the freeze, so an event on the final day has $0 \leq \Delta t < 1$ and one on the first day is 365 days old. The registered analysis (Appendix 8.6) scores every event of scan A from 2023-10-01.

- BigQuery source: Logs are read from the public BigQuery table

  `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`.

  One record is one event emitted by a contract, with its block timestamp, address, topics and data. Scans A and B copy the events the study needs into its own Google Cloud project, and later queries read those copies (Section 3.3). Anyone with a Google Cloud account can rerun the same queries against the same public table; Appendix 8.2 shows the query patterns.

- Reputation events: EndorseRank uses ERC-20 `Approval` logs (Vogelsteller & Buterin, 2015), which also cover approvals made through EIP-2612 `permit` (Lundfall, 2020), and AWP uses `Transfer` logs. Both are kept only for logs with exactly three topics and for the five tokens of Section 3.5.1. GMX V2 position closes supply the trader labels.

- Scored addresses: Only contract accounts are scored. An externally owned account (EOA), the wallet of an individual user, can serve for trading, holding, payments, bridging, airdrop farming or an automated bot, and users mix these uses in proportions that differ from one to the next, so EOAs have no common pattern of activity on which they could be put in one order. A contract's role is fixed by its code (a router, a vault, a pool), and the owners that approve it grant the same kind of permission for that role, so the approvals and transfers that different contracts receive can be compared. EOAs therefore enter the graphs only as owners, senders and recipients. That comparability holds within a role, however. The cohort mixes roles and can include contracts that individuals use as wallets, and no analysis is stratified by role, so differences between roles enter every coefficient. An address counts as a contract when `eth_getCode` at block $513{,}216{,}506$ (9 October 2026) returns code other than an EIP-7702 delegation designator, which an EOA can set while keeping its key (Buterin et al., 2024); addresses outside the cohort candidates were read at later blocks within three hours (Appendix 8.1.3).

- Matched cohort: Alignment is applied to the $n = 6{,}483$ contracts that received a non-zero `Approval` of one of the five tokens from at least three distinct owners inside the score window of T_{obs}, the part of the plan's cohort ($27{,}844$ contracts approved on any token from 2023-10-01) that meets the rule on the five tokens within that window. W0 draws a cohort of its own by the same rule inside its own score window, so that no event after t_1 chooses the contracts around which its data are drawn (Section 3.7). The list is fixed from the approvals and the code check alone, but it is not independent of the allowance graph: every cohort contract held allowances at some time in the window. A cohort contract need not appear in either graph; the method concerned then scores it 0 (Section 3.6), and Section 4.1.1 counts these contracts.
- Ego network: Only events in which a cohort contract is owner or spender, or sender or recipient of a non-zero transfer, are used. The extraction keeps those of the plan's cohort, and each freeze keeps those of its own cohort in its score window and its label window. This bounds the extracted volume and makes each graph a one-hop ego network around the cohort, since an event between two addresses outside it is never used. An address outside the cohort, such as a holdout spender that fewer than three owners approved or almost any GMX trader, is therefore seen only through its events with cohort contracts: its degrees, activeness and scores, and a spender's labels, count only those. Endorsement passes over more than one hop only along edges that each touch a cohort contract; Section 5.3.2 examines how far the allowance walk does so.
- Exclusions: Mempool transactions, off-chain order books, centralized exchange accounts and bridges lie outside the data, and no attempt is made to merge addresses controlled by one operator into one entity. Tokens other than the five are left out, and native ether enters only as WETH.

3.3 Two-phase collection strategy

Data collection and evaluation run as two separate phases, so the tables, benchmarks and robustness runs can be rebuilt as often as needed without cloud costs, by us or by anyone who has run the extraction.

- Phase 1 (reputation streams): Scan A copies the approval, transfer and GMX events from 1 October 2023 to 30 June 2026 into the study's project, and scan B those of July to September 2026 once the registration of W1 is committed (Section 3.7.2). One query lists the spenders approved by at least three owners, the code check fixes the plan's cohort, and a further query extracts its ego events. From these the main analysis keeps the five tokens, draws the cohort of each freeze from the approvals of its score window and keeps the events of that cohort. At each anchor date, T_{obs} and t_1, the approvals of the anchor's score window are reduced to the most recent allowance per (token, owner, spender) triple in (block number, log index) order and kept when positive, the transfers of the window are aggregated per sender–recipient pair, and amounts are valued in US dollars (Section 3.5.1); the proxies and holdout labels are computed alongside. These tables are the only inputs to graph construction.
- Phase 2 (local evaluation): From the downloaded Phase 1 tables a local machine decodes the GMX closes and computes the scores, the alignment statistics, the holdouts, the benchmarks and the robustness runs, without querying BigQuery.
- Cost guardrails: Every query is first run as a dry run, and a query whose estimate would take the spending past the study's budget of USD 50 is aborted unless the operator confirms it with an extra flag.[2] Every job is written to a cost ledger with its billed bytes and cost (Appendix 8.1), so the cost of acquisition can be audited.

[2]This budget is enforced by the study's own pipeline; BigQuery does not impose it.

3.4 Pipeline architecture

Figure 3.2 traces the whole pipeline, from public logs to the research tables. All remote steps write tables in the study's BigQuery project, and the downloaded files are never changed.

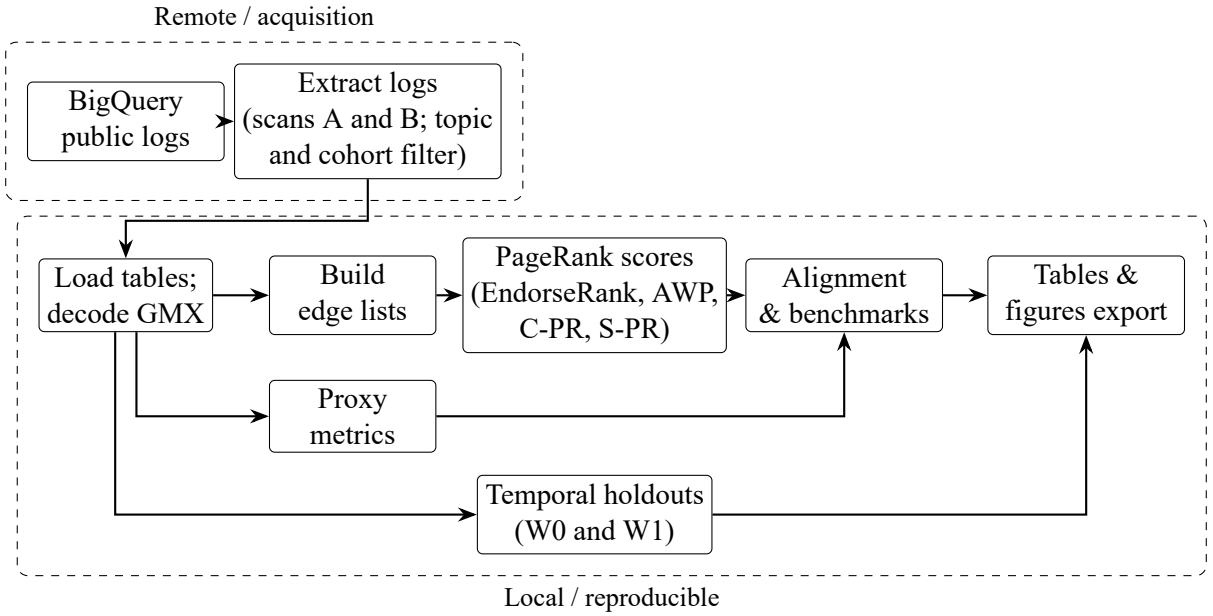


Figure 3.2: Pipeline overview. Approval, transfer and GMX logs are copied remotely in two scans and aggregated into pair tables for the cohort. Locally, the GMX closes are decoded and the pair tables turned into edge lists, from which EndorseRank, AWP, C-PR and S-PR are computed; the proxy metrics come from the same events on a parallel branch. The alignment and benchmark calculations and the temporal holdouts follow, and the result tables and figures are exported last.

Unit tests run the scoring and statistics code on synthetic fixtures.

3.5 Event decoding

A log emitted by a smart contract has topics and a data payload but no field names. Decoding turns each row into a typed record for graph construction and the checks, and an error here would corrupt both graphs silently. The topic hashes and field mappings are listed below.

- Log anatomy: Each log records the emitting contract, a sequence of topics and an ABI-encoded data blob. The first topic is the keccak-256 hash of the event signature; indexed

parameters follow as 32-byte words (addresses left-padded), and non-indexed parameters go into the data field in ABI order (Vogelsteller & Buterin, 2015). ERC-721 tokens emit `Transfer` and `Approval` under the same signatures but put the token ID in a fourth topic, so the condition of exactly three topics keeps them out. SQL filters by topic and decodes the ERC-20 fields, and Python decodes the GMX events.

- ERC-20 `Approval`: Its topic hash is

  `0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925`,

  the keccak-256 hash of `Approval(address,address,uint256)`. Owner and spender occupy Topics 1 and 2, and the allowance is decoded from the data field in the token's base units and valued in US dollars (Section 3.5.1). An allowance of zero is kept, since it resets the allowance in force. A `permit` emits the same `Approval` event, so an allowance counts whether it came from an on-chain `approve()` call or from an off-chain signature (Lundfall, 2020).

- ERC-20 `Transfer`: Its topic hash is

  `0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef`,

  the keccak-256 hash of `Transfer(address,address,uint256)`. Sender and recipient are indexed, and the amount is valued in US dollars at the price of its day. Zero-value transfers and transfers made before their token's first price are dropped. Mint and burn logs, which use the zero address, are kept whenever a cohort contract is involved. Self-transfers stay in the events table but are left out of the inbound-counterparty computations (Section 3.9), since they reveal no counterparty.

- GMX V2 position closes: A close is one `PositionDecrease` log of the GMX V2 event emitter, so partial closes and liquidations count. Python decodes its account, base profit or loss and order type; every close of a contract account decoded without error.

- Address normalization and units: Addresses are lower-cased before the join. Amounts are the `uint256` value as a double-precision float, divided by 10^q for a token with q decimals and

multiplied by its US-dollar price, so an unlimited allowance, $2^{256} - 1 \approx 1.16 \times 10^{77}$ base units, becomes a very large dollar amount (Sections 3.6.2 and 3.6.3 give its effect).

3.5.1 Tokens, prices and the revision of the amounts

The analysis plan fixed amounts in raw base units, every ERC-20 token and $b = 1$ in Equation (3.4). That setting has three defects. First, $V(z)$ is within 10^{-4} of 1 for any amount of 10 base units or more, which almost every transfer and allowance reaches, so the amount played no part in EndorseRank or AWP. Second, the layers of C-PR added raw base units across tokens with different decimals, and an unlimited allowance took almost the whole row of every owner that held one. Third, any token contract can emit `Approval` and `Transfer` logs naming an owner or sender that took no part, so the log of an arbitrary token is no evidence of an endorsement or a payment.

The revised analysis was therefore defined after the registered results were known. Its amounts are in US dollars, it keeps five tokens, and it scores each contract on the twelve months before each freeze (Section 3.2). A token qualifies when Chainlink's directory of data feeds on Arbitrum One lists a market-price feed for it, or for the asset it represents, with a low or medium market-risk category (Chainlink, n.d.), and a qualifying token is priced when DefiLlama has a daily price for it (DefiLlama, n.d.). Of the 133 tokens the directory maps, 64 qualified and 46 of these are priced; the analysis keeps the five priced tokens with the most rows in the score window of T_{obs}: WETH, USDC, USDT0, WBTC and ARB. A transfer is valued at the DefiLlama daily close of its UTC day, and a latest allowance at the price of the anchor's day. The slope of V is $b = \ln 3/m$, so that an amount of m dollars weighs one half; m is the median US-dollar amount in the W0 score window, so that no event after t_1 enters it: $m_T = 170$ dollars for transfers and $m_A = 100$ dollars for the finite latest allowances at t_1. In the allowance layer of C-PR an allowance counts at most $U = 19{,}000$ dollars, the 99th percentile of single transfers in the same window. The 99th percentile of the finite allowances themselves is about 10^{63} dollars, so a cap taken from them would leave unlimited allowances in charge of the rows. The cohorts are drawn on the five tokens from the ego events already extracted, without a new scan.

Rules A and B and the comparison on outcomes built from neither edge type are applied to the revised scores with their thresholds and wording; since these scores were defined after the registered results were known, the verdicts are a re-analysis and not registered tests. Appendix 8.6 sets the registered analysis beside the revised one, and Appendix 8.4.2 gives the tokens, the rows left out and the price checks.

3.6 Graph construction

EndorseRank and AWP weight their edges the same way and run the same PageRank solver, Equation (3.1); they differ in the events that make the edges and in where the walk restarts. Each method keeps the edges with at least one endpoint in the seed set $\mathcal{W}$, which brings in their one-hop neighbors as nodes, and reports a score for every evaluated contract; a contract missing from the subgraph scores 0.

Edges in EndorseRank stand for endorsements and edges in AWP for transfers (Section 2.7). Both methods weight an event by its age, so that recent events count for more, and pass its US-dollar amount through a bounded function, so that a larger amount counts for more but one event counts at most once. AWP is described first, since EndorseRank takes its weights from it, and C-PR, which walks both edge types, comes last.

3.6.1 AWP transfer graph

AWP is the method of T. Do et al. (2023), presented at IEEE RIVF 2023. A transfer e worth x_e US dollars, made Δt_e days before the score window closes at its anchor (T_{obs}, or t_1 in W0), enters the edge from its sender to its receiver with the weight $\sigma(\Delta t_e)\, V(x_e)$. The time weight is the logistic decay

$$\sigma(\Delta t) = \frac{1}{1 + \exp\big(k(\Delta t - t_0)\big)}, \qquad k = 0.01, \quad t_0 = 180 \text{ days}, \tag{3.3}$$

and the value weight is

$$V(z) = \frac{2}{1 + e^{-bz}} - 1, \qquad b = \frac{\ln 3}{m}, \tag{3.4}$$

which is 0 at $z = 0$, $1/2$ at the median amount $z = m$ and rises towards 1 as the amount grows. For transfers $b = b_T = \ln 3/m_T$ (Section 3.5.1). The edge weights are sums over transfers:

$$w_T(u, v) = \sum_{e:\, u \to v} \sigma(\Delta t_e)\, V(x_e). \tag{3.5}$$

The sum runs over every transfer from u to v in the window, whatever the token. A self-transfer ($u = v$) is kept and becomes a self-loop on u. The walk restarts at an address in proportion to its activeness,

$$X_u = \sum_{v \neq u} \max_{e:\, u \to v} \sigma(\Delta t_e), \tag{3.6}$$

which adds up, over the addresses that u has sent to, the time weight of its latest transfer to each. The restart distribution of Equation (3.1) is $\pi_u = X_u / \sum_{v \in V_T} X_v$, and it also takes the mass of the dangling nodes. An address that never sends to another has $X_u = 0$ and receives no restart mass, so its score comes from its in-edges alone. The damping factor is $d = 0.85$.

The paper scores Ethereum transactions and gives no parameter values; its time weight is the logistic function of Equation (3.3) written in calendar time instead of age, and k, t_0 and b are this study's choices. A transfer of the median amount weighs one half and one of twice the median 0.8, so an edge weight grows with the amounts transferred as well as with their number. Within a score window the decay discounts older events strongly (Figure 3.3): the weight falls from $\sigma(0) = 0.858$ through $\sigma(182) = 0.495$ to $\sigma(365) = 0.136$ for an event from the first day of the window.

Algorithm 1 states the construction and scoring steps formally.

3.6.2 EndorseRank allowance graph

EndorseRank puts AWP's edge weights on the allowance graph. Let $\mathcal{A}$ denote the set of latest allowances after preprocessing. For each (token, owner–spender) triple only the most recent `Approval` event inside the score window is kept, in (block number, log index) order, and a zero allowance deletes the edge. The surviving allowance $a^\star(u, v, \text{token})$, in US dollars, was set $\Delta t^\star(u, v, \text{token})$

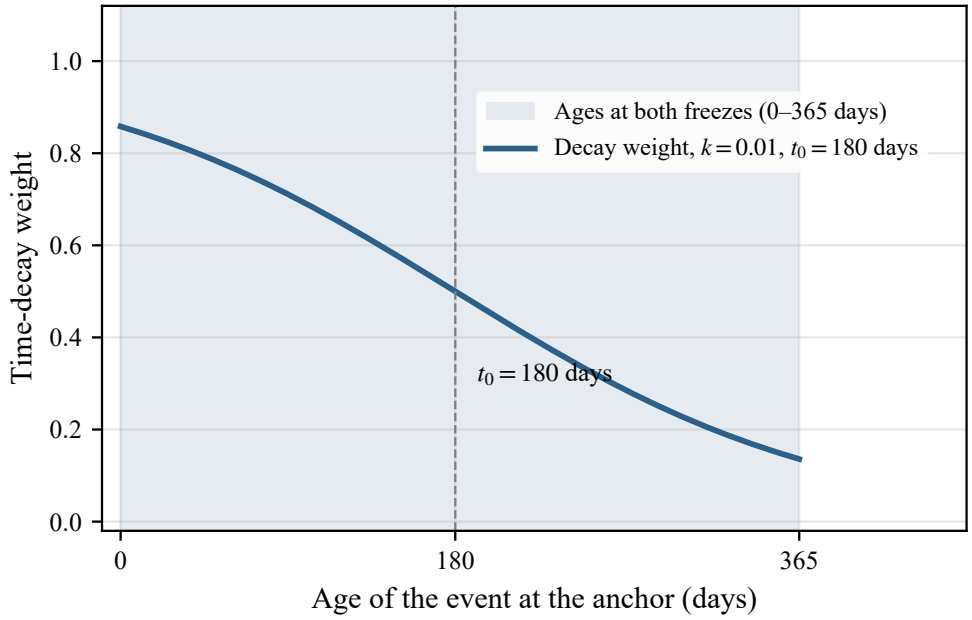


Figure 3.3: Logistic time-decay $\sigma(\Delta t)$ of AWP and EndorseRank ($k = 0.01$, $t_0 = 180$ days). The horizontal axis shows the age of an event in days before the anchor, $T_{\text{obs}} = 30$ June 2026 or, in W0, $t_1 = 31$ March 2026, and the vertical axis shows the time weight of a transfer or an approval. The shaded band marks the ages that occur at either freeze (0 to 365 days), over which the weight falls from 0.86 to 0.14.

Algorithm 1 AWP graph construction and scoring

Require: Transfers $\mathcal{T}$ in US dollars, seed contracts $\mathcal{W}$, observation end T_{obs}, parameters $k, t_0, b_T, d, \varepsilon, I_{\max}$

Ensure: Scores $\mathbf{s}_{\text{AWP}}$ on $\mathcal{W}$

$E \leftarrow \emptyset$; $X_u \leftarrow 0$ for every address u
for each transfer $e = (u, v, x_e, t_e)$ in $\mathcal{T}$ **do**
    $\Delta t \leftarrow (T_{\text{obs}} - t_e)$ in days
    accumulate $\sigma(\Delta t)\, V(x_e)$ on edge $(u \to v)$ ▷ Equations (3.3)–(3.5); $u = v$ gives a self-loop
end for
for each sender u **do**
    $X_u \leftarrow \sum_{v \neq u} \max_{e:\, u \to v} \sigma(\Delta t_e)$ ▷ Equation (3.6)
end for
remove non-positive edges; retain edges incident to $\mathcal{W}$
$\pi_u \leftarrow X_u / \sum_v X_v$ for the nodes u of the retained edges
$\mathbf{s} \leftarrow \text{WeightedPageRank}(E, \boldsymbol{\pi}, d, \varepsilon, I_{\max})$ ▷ Equations (3.1)–(3.2)
return $\mathbf{s}_{\text{AWP}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{W}$ (else 0)

days before T_{obs}, and it enters the edge as one event of that age and amount:

$$w_A(u,v) = \sum_{\text{token}} \sigma\big(\Delta t^\star(u,v,\text{token})\big)\, V\big(a^\star(u,v,\text{token})\big), \tag{3.7}$$

with σ, V, k and t_0 as for AWP and $b = b_A = \ln 3/m_A$. A directed edge $u \to v$ means that owner u endorses spender v. An allowance of the median amount enters with half its time weight and a large or unlimited one with close to its full time weight, so $w_A(u,v)$ grows with the amounts that u has approved to v, but no single allowance counts for more than one. The damping factor is $d = 0.85$. The walk restarts uniformly. AWP's restart rule, carried over to allowances, restarts the walk at owner u in proportion to $X_u^A = \sum_{v \neq u} \max_{\text{token}} \sigma\big(\Delta t^\star(u,v,\text{token})\big)$, summed over the spenders v of u's positive latest allowances, a weight that grows with the number of spenders u approved recently. That rewards granting endorsements, while EndorseRank is meant to rank the contracts that receive them. The analysis plan fixed uniform restarts and named this variant, EndorseRank with activity restarts, as one to report next to EndorseRank (Table 4.8).

The weight keeps the amount and the time of the latest approval; it is not the allowance in force at the end of the window. Within a pair, on each token, the newest approval replaces the older ones (Vogelsteller & Buterin, 2015). An allowance last set before the score window began is not an edge, even if it is still in force, so the allowance graph holds the authorizations that owners granted or renewed in the year before the freeze. An owner who approves an allowance-manager contract such as Uniswap's Permit2 appears only with an edge to that contract, not to the application that later spends through it. Some tokens also emit `Approval` from inside `transferFrom` with the reduced allowance. For them the latest record can be the remainder after the spender's last use, dated at that spend, so the time weight partly records use, an allowance spent in full ends at zero and drops out, and an authorization granted before the score window and used in a label window can appear as a new approval pair. For the other tokens a reduced allowance keeps its approved value. How often the latest record is such a remainder was not measured.

The steps for building the graph and running PageRank are listed in Algorithm 2.

Algorithm 2 EndorseRank graph construction and scoring

Require: Latest allowances $\mathcal{A}$ in US dollars, seed contracts $\mathcal{W}$, observation end T_{obs}, parameters $k, t_0, b_A, d, \varepsilon, I_{\max}$
Ensure: Scores $\mathbf{s}_{\text{ER}}$ on $\mathcal{W}$

$E \leftarrow \emptyset$
for each latest allowance $(u, v, \text{token}, a^\star, t^\star)$ in $\mathcal{A}$ **do**
    $\Delta t^\star \leftarrow (T_{\text{obs}} - t^\star)$ in days
    accumulate $\sigma(\Delta t^\star)\, V(a^\star)$ on edge $(u \to v)$ ▷ Equation (3.7)
end for
remove non-positive edges; retain edges incident to $\mathcal{W}$
$\mathbf{s} \leftarrow \text{WeightedPageRank}(E, \text{uniform } \boldsymbol{\pi}, d, \varepsilon, I_{\max})$ ▷ Equations (3.1)–(3.2)
return $\mathbf{s}_{\text{ER}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{W}$ (else 0)

Approvals are rarer than transfers, so on the matched cohort the approval subgraph is sparser (Chapter 4); since one PageRank iteration costs $O(|E|)$, this sparsity is the structural basis of H3.

For both methods, edge lists are stored as (from, to, weight) triples; duplicate edges are merged by summing their weights, and non-positive weights are dropped before the solve.

3.6.3 Coupled operator over both layers (C-PR) and the seeded ablation (S-PR)

EndorseRank and AWP each walk on a single edge type, so each ranks a contract by one kind of act: the spending rights it is granted or the payments it receives. C-PR, the method of the thesis, walks both edge types of the same addresses in one chain. RQ4 asks whether it ranks contracts better than a walk over either layer alone, and RQ5 whether it improves on a walk over the transfers alone. The analysis plan fixed the coupled operator on two layers that weight each event by its amount, which the revision values in US dollars (Section 3.5.1):

$$\tilde{w}_A(u,v) = \sum_{\text{token}} \min\big(a^\star(u,v,\text{token}), U\big), \qquad \tilde{w}_T(u,v) = \sum_{e:\, u \to v} x_e\, \sigma(\Delta t_e). \tag{3.8}$$

The allowance layer has the edges of EndorseRank and the transfer layer those of AWP, apart from the pairs whose value weight is zero in floating point (Section 4.1.1), which only the layers of C-PR keep; otherwise only the weights differ. Amounts enter with no bounded transform, an allowance

counts at most U, and the allowance layer carries no time decay. Both rules for C-PR were fixed on these layers in raw base units. The two layers form one graph on the union node set $V = V_A \cup V_T$, with transition matrices P_A and P_T from Equation (3.2). For node u the coupled row is a convex mixture of the two layer rows, renormalized over the layers in which u has out-edges:

$$P^{(\lambda)}_{u,\cdot} = \alpha_A(u)\, P_{A,u,\cdot} + \alpha_T(u)\, P_{T,u,\cdot}, \tag{3.9}$$
$$\alpha_A(u) = \frac{\lambda\, \mathbf{1}[o_A(u) > 0]}{D_\lambda(u)}, \qquad \alpha_T(u) = \frac{(1-\lambda)\, \mathbf{1}[o_T(u) > 0]}{D_\lambda(u)},$$

where $D_\lambda(u) = \lambda\, \mathbf{1}[o_A(u) > 0] + (1-\lambda)\, \mathbf{1}[o_T(u) > 0]$, and $o_A(u)$ and $o_T(u)$ are the out-strengths of u in the two layers. A node with out-edges in only one layer follows that layer with weight one. A layer of weight 0 is dropped with its edges and nodes, and a node with $D_\lambda(u) = 0$ is dangling, its mass handled by the third term of Equation (3.1). Teleportation is uniform, $d = 0.85$, and $\mathbf{s}_{\text{CPR}}$ is the PageRank vector of Equation (3.1) with $P = P^{(\lambda)}$, under the solver settings of the single-layer methods. Figure 3.4 shows one step of the walk, and Algorithm 3 lists the construction and scoring steps.

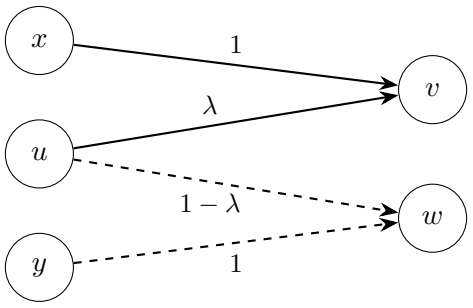


Figure 3.4: One step of the coupled walk C-PR (Equation (3.9)); solid arrows are allowance edges and dashed arrows transfer edges, labelled with their transition probabilities before damping. The address u has out-edges in both layers and mixes them with weights λ and $1-\lambda$; x and y have out-edges in one layer only and follow it whatever λ is. With probability $1-d$ the walk restarts at a uniformly chosen address.

At $\lambda = 1$ the operator walks the allowance layer alone and at $\lambda = 0$ the transfer layer alone; we write these walks C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$). They are the comparators of both rules for C-PR (RQ4 and RQ5; Sections 3.7 and 3.7.2). Unit tests check on a fixture that each end point gives

Algorithm 3 C-PR graph construction and scoring

Require: Latest allowances $\mathcal{A}$ and transfers $\mathcal{T}$ in US dollars, seed contracts $\mathcal{W}$, observation end T_{obs}, cap U, weight λ, parameters $k, t_0, d, \varepsilon, I_{\max}$

Ensure: Scores $\mathbf{s}_{\text{CPR}}$ on $\mathcal{W}$

allowance layer: accumulate $\min(a^\star, U)$ on $(u \to v)$ for each latest allowance in $\mathcal{A}$ ▷ Equation (3.8)

transfer layer: accumulate $x_e\, \sigma(T_{\text{obs}} - t_e)$ on $(u \to v)$ for each transfer in $\mathcal{T}$

remove non-positive edges; retain edges incident to $\mathcal{W}$; drop a layer of weight 0

$V \leftarrow V_A \cup V_T$; $P_A, P_T \leftarrow$ row-normalized layers ▷ Equation (3.2)

for each node $u \in V$ with $D_\lambda(u) > 0$ **do** ▷ the others are dangling

  $P^{(\lambda)}_{u,\cdot} \leftarrow \alpha_A(u)\, P_{A,u,\cdot} + \alpha_T(u)\, P_{T,u,\cdot}$ ▷ Equation (3.9)

end for

$\mathbf{s} \leftarrow \text{WeightedPageRank}(P^{(\lambda)}, \text{uniform } \boldsymbol{\pi}, d, \varepsilon, I_{\max})$ ▷ Equation (3.1)

return $\mathbf{s}_{\text{CPR}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{W}$ (else 0)

the scores of its layer walked alone with uniform restarts. The family is not continuous at the end points. For $0 < \lambda < 1$ the node set is $V_A \cup V_T$ and a node with out-edges in only one layer follows it whatever λ is, whereas at $\lambda = 1$ a node whose out-edges are all transfers is dangling or absent; as λ approaches 1, C-PR therefore tends to a walk that still follows the transfer edges of such nodes, and the same holds at the other end. The values $\lambda = 0.5$ (primary) and $\lambda \in \{0.25, 0.75\}$ (sensitivity) were fixed in the plan's configuration.

Uncapped amounts would make the allowance layer very uneven: close to half of the positive latest allowances at T_{obs} are unlimited (Appendix 8.4.2), each would take almost the whole row of its owner, and the finite allowances of the owners who hold both kinds would have almost no effect. With the cap an unlimited allowance weighs as much as a finite allowance of U dollars or more, and the smaller allowances keep a share of the row in proportion to their amounts. EndorseRank and AWP bound every amount through V instead.

In the S-PR ablation the transfer layer is walked, but the uniform teleportation vector $\boldsymbol{\pi}$ of Equation (3.1) is replaced by the score of C-PR ($\lambda = 1$), restricted to V_T and renormalized over it. Let $\mathbf{r}_A$ be the PageRank vector of the allowance layer walked alone on V_A, extended by $r_{A,u} = 0$ for

$u \notin V_A$. Then

$$\pi_u = \frac{r_{A,u}}{\sum_{v \in V_T} r_{A,v}} \qquad (u \in V_T),$$

so a transfer-graph node without an allowance score receives zero restart mass. If no node of V_T lies in V_A, the denominator is zero and $\boldsymbol{\pi}$ falls back to the uniform distribution on V_T. The same vector redistributes dangling mass. The result is the TrustRank construction (Gyöngyi et al., 2004), but with allowance-derived seeds instead of hand-labeled ones. Allowances decide where the walk restarts; transfers decide how it spreads. A comparison of S-PR with C-PR separates what the allowance edge contributes as a propagation medium from what it contributes as a seed. Like AWP and C-PR ($\lambda = 0$), however, S-PR scores zero every contract without a transfer edge, about one spender in six at each freeze (Section 4.6). Allowance edges order these spenders, so any contrast of a transfer-only walk with a score that orders them, F1 of rule B included, mixes ranking with coverage.

The edge set determines what the coupled operator costs. Because C-PR walks on $E_A \cup E_T$, one of its iterations costs as much as an iteration on each layer, and its runtime is expected to be at or above AWP's. The speed result of Claim C5 is therefore EndorseRank's alone. S-PR needs one solve on each layer. Both operators are timed under the benchmark protocol of Section 3.11.

3.7 Temporal holdout protocol

In the holdout W0, scores use the events of the W0 score window, from 1 April 2025 up to $t_1 =$ 31 March 2026, 23:59:59 UTC. Labels are counted on events from 1 April 2026, 00:00:00 UTC, up to $t_2 =$ 30 June 2026, 23:59:59 UTC. The cohort consists of the $n = 7{,}914$ contract spenders that hold at least one positive latest allowance of one of the five tokens, set inside the window, at t_1. These data are the ego network of the W0 cohort, which the rule of Section 3.2 draws from the approvals of the same window, so no event after t_1 chooses the contracts around which the W0 data are drawn. Every score in the holdout table (EndorseRank, AWP, C-PR at the three values of λ and at its two end points, S-PR and two raw-degree baselines) is computed from these events

alone. The time decay is anchored at t_1 as well: Δt in Equation (3.3) is measured back from t_1, so ages run up to 365 days and the weights from 0.858 down to 0.136.

We count two labels per spender, one for each construct, so that in the future window each single-layer method is tested on its own construct and the hybrids on both:

1. New approval pairs: the count of owner–spender pairs, attributed to the spender, that receive a positive approval in the label window but were absent from the latest-positive snapshot at t_1. This label measures the endorsement construct.
2. New transfer senders: the count of distinct addresses that send the spender a transfer in the label window without having sent it one between the start of the score window (1 April 2025) and t_1, self-transfers excluded. It measures the flow construct and is the transfer-side counterpart of the first label.

For baselines we take each spender's in-approve degree (the number of distinct owners with a positive allowance) and its transfer in-degree (the number of distinct senders other than the spender itself), both at t_1. Each single-layer PageRank smooths one of these local statistics. Because their rows appear in the table, the increment of each PageRank over its own raw degree can be read off directly. That increment answers RQ3; the PageRank coefficient on its own does not. C-PR smooths both statistics, so it is read against both.

Uncertainty comes from the paired bootstrap over contracts of Section 3.10 ($B = 400$, one shared index matrix), so every difference between two coefficients on the same label is a paired difference. The contrasts fixed in the plan are the two of rule A below. Every other contrast in this window, each PageRank against its own degree and those of S-PR among them, was computed after the labels were known; the contrasts of C-PR with every other score are in Table 4.18, and the differences between EndorseRank and AWP in Appendix 8.5.

Rule A, fixed in the analysis plan, judges C-PR against its own layers walked alone and has two parts. The primary part uses W0: C-PR at $\lambda = 0.5$ should do no worse than C-PR ($\lambda = 1$) on new approval pairs (the interval of $\Delta\tau = \tau_{\text{CPR}} - \tau_{\lambda=1}$ contains zero or lies above it) and should be

better than C-PR ($\lambda = 0$) on new transfer senders (the interval of $\tau_{\text{CPR}} - \tau_{\lambda=0}$ lies above zero). The secondary part uses the same window and the matched cohort. Its condition is that C-PR's point estimate lies within the 95% interval of the better of the two layers on the transfer family and on the allowance family, and that C-PR exceeds both layers on the Sybil-stability family. The paired contrasts of Table 4.14 are reported beside this rule as a stricter check; they do not decide it. If its primary part fails, RQ4, whether C-PR beats a walk on either layer alone, receives a negative answer, and the result is reported with its intervals. RQ5, whether C-PR improves on a walk over the transfers alone, is not covered by rule A; rule B tests it in W1 (Section 3.7.2). Once labels have been observed, λ is not tuned. When rule A was fixed, the study had extracted no log; its verdict on the revised scores is a re-analysis (Section 3.5.1).

The pipeline also counts, over the label window, revocations, the pairs held at t_1 that receive a zero approval (for some tokens an allowance spent in full, Section 3.6.2), and an approval-then-outbound drain heuristic. Neither is offered as supporting evidence (Section 4.6).

3.7.1 Trader outcomes in W0

The spender cohort says little about contracts that trade. An exploratory run applied the freeze t_1 to the 87 GMX V2 accounts that are contracts, appear in the freeze-date graph and close at least three positions from April to June 2026, with success labels from that window: the number of profitable closes, the realized gain (the sum of the positive base profits of the closes, losses not subtracted) and the shares of closes that were profitable and that were not losses; a liquidated close counts as not profitable and, since the pipeline does not count it as a loss, as not a loss. Every score of the spender holdout was set against these labels under no rule, with 400 paired resamples, and Chapter 4 reports the run as exploratory. It has no liquidation label, and with so few contract traders every interval is wide.

3.7.2 Registered window W1: July to September 2026

The W0 holdout also showed something that no rule had asked about. Against its transfer layer walked alone, C-PR ranked the spenders that later gained new approval pairs higher, with an interval above zero, while it was worse on new transfer senders (in the registered analysis; Appendix 8.6). We noticed this only after the labels were in, so it counts as a hypothesis and not as a finding, and it is tested again on a window that the study had not extracted when the test was fixed. The registration document, written after the W0 results were known, and the configuration, with the status of the window set to registered, were committed to the authors' Git repository at 16:49 UTC on 9 October 2026 (commit `1747e64`), and scan B, the only query that reads logs dated after 30 June 2026, ran after that commit. The extraction script refuses to run unless both files are committed without local changes and the status is registered, and it records the commit and the hashes of both files in an extraction manifest (Appendix 8.1).

The method is unchanged. C-PR is used at $\lambda = 0.5$, as fixed in the plan, together with the two sensitivity values and S-PR. The solver settings, the SQL, the extraction filter of the matched cohort, the GMX decoder and the paired bootstrap (400 resamples, seed 42) are those used for W0. Scores are frozen at T_{obs}, 30 June 2026 at 23:59:59 UTC, so they are the scores of the same-window evaluation: in the registration, which belongs to the registered analysis, they use every extracted event from 1 October 2023 onwards, and in the revised analysis the events of the score window of T_{obs} (Section 3.2). Labels are counted from 1 July to 30 September 2026.

Two cohorts are evaluated, and Section 4.1.1 gives their sizes. The spender cohort is defined as in W0: every contract spender holding at least one positive latest allowance at the freeze. It holds about five in six of the W0 spenders, so W1 is a later label window on largely the same contracts. The trader cohort is the GMX V2 accounts that are contracts, are nodes of the freeze-date graph in either layer and close at least three positions in the label window. The spenders keep their two labels. The traders get the liquidation-free close rate, one minus the share of a contract's July–September closes that ended in liquidation. It is the nearest thing to a default that these data contain, it is built

from neither graph, and because it is a rate it does not reward a contract merely for trading more. When it was chosen, no score of the study had been set against a liquidation label; the study knew only, from the decoding step, how common liquidations are among the closes of contract accounts (Section 4.1.1).

Rule B compares C-PR with its transfer layer walked alone, C-PR ($\lambda = 0$), through the paired difference $\Delta\tau = \tau_{\text{CPR}} - \tau_{\lambda=0}$:

- F1, spenders, new approval pairs: the 95% interval of $\Delta\tau$ lies above zero;
- F2, spenders, new transfer senders: the lower end of the interval lies above $-\delta$;
- F3, traders, liquidation-free close rate: the lower end of the interval lies above $-\delta$.

C-PR counts as improving on a walk over the transfers alone only if all three hold, so no correction for multiple testing is needed. The margin $\delta = 0.01$ is the half-width of the paired interval for the F2 contrast in W0 in the registered analysis, rounded up to two decimals, as the registration fixed it (Appendix 8.6). In the registered analysis the spender cohorts could resolve differences smaller than the margin, so it was wider than those data required; on the smaller cohorts of the revised analysis the paired interval of F2 is about as wide as the margin (half-width 0.012 in W1), and on the traders the lower end can clear $-\delta$ only if the two walks rank them almost alike, so F3 had little power to pass. One weakness of F3 was written down with the rule. If neither score predicts liquidation, F3 can pass with both coefficients near zero, so the two coefficients are reported next to the verdict. Three further results are reported in a fixed order and do not enter the decision: C-PR against C-PR ($\lambda = 0$) on the liquidation-free close rate as a superiority test (F4), C-PR against C-PR ($\lambda = 1$) on the same label (F5), and the primary part of rule A, rerun on the new window (F6).

Two sensitivity analyses were registered in the same commit, before the study extracted any July–September log. Both are reported next to the decision, and neither enters it. The first repeats F1–F4 with AWP (Section 3.6.1) in place of C-PR ($\lambda = 0$). The second repeats F1–F6 with the cohort contracts that a graph leaves out kept as isolated nodes instead of scored zero (Section 4.4). Only the

two layers walked alone change under the second analysis, since every spender is in the allowance graph and every trader in the union graph that C-PR walks. Other contrasts in W1, each PageRank against its own degree and C-PR against every other score among them, were computed after the labels were known and are reported apart from the decision.

3.7.3 Outcomes built from neither edge type

A comparison of the two edge types on the GMX trader labels, which are built from neither, was registered in the same commit as rule B. As in a preregistration (Nosek et al., 2018), its contrasts, rules and wording were written down before the study extracted the July–September logs and before any W0 liquidation label was computed; on the revised scores they are applied as a re-analysis. It reuses the trader cohorts and label code of Sections 3.7.1 and 3.7.2 in two windows: W1 (primary), frozen on 30 June 2026 with labels from July to September (70 traders), and W0 (secondary), frozen on 31 March 2026 with labels from April to June (87 traders).

Four pairs set allowance against transfer: in-approve degree and transfer in-degree; EndorseRank with AWP's activity restarts and AWP, the like-for-like PageRank pair; EndorseRank and AWP; and the amount-weighted layers C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$). C-PR ($\lambda = 0$) minus AWP isolates AWP's weights and restarts. All six trader labels are reported: the liquidation-free close rate (primary), a no-liquidation flag, profitable closes, realized gain, and the profitable and non-loss shares of closes. Two contrasts on the liquidation-free close rate in W1 decide: P, in-approve degree minus transfer in-degree, and Q, EndorseRank with activity restarts minus AWP, each with a Bonferroni 97.5% interval from $B = 2{,}000$ paired resamples (seed 42). Because the chance of at least one liquidation rises with the number of closes, τ_b is also computed within strata of 3–4, 5–9, 10–24 and 25 or more closes in the label window and weighted by each stratum's contract pairs. Recipients (traders with an in-approve degree above zero at the freeze) are compared with the other traders on every label, and an influence check drops the 10 recipients of highest in-approve degree. Every W1 recipient and 38 of the 39 W0 recipients were approved by one and the same cohort contract, so the recipients form one cluster. The two trader cohorts share 56 contracts, all

W1 recipients among them, so W0 is not an independent second sample.

R1, a count-level lead on liquidation avoidance, needs the 97.5% interval of P and the 95% interval of the stratified P above zero in W1, and a positive P in W0. R2, a PageRank-level lead, needs the 97.5% interval of Q above zero in W1 and a positive Q in W0. R3 applies if P on the profitable or the non-loss share in W1 is below zero with a 95% interval excluding zero; the allowance signal is then read as specific to liquidation avoidance. The wording of the conclusion for each outcome was fixed in the same commit.

3.8 Construct-check design

Every score enters the same-window evaluation: EndorseRank, AWP, C-PR at its five values of λ and S-PR. For each score M and check q the pipeline reports Spearman's ρ and Kendall's τ between $\mathbf{s}_M$ and $\mathbf{p}_q$ with 95% bootstrap intervals (Section 3.10), each check oriented so that a higher value means higher expected reputation, as in the AWP evaluation protocol (T. Do et al., 2023).

The same-window evaluation is organized around two contemporaneous families:

1. Transfer (the proxies used to evaluate AWP (T. Do et al., 2023)): the contract's inbound transfer activity.
2. Allowance: inbound approval activity, where the contract is the spender.

Both families are graph-adjacent: their inputs are the event classes that make up G_{transfer} and G_{approve}, although each check is a local statistic and not a global score. Alignment of AWP with the transfer checks, or of EndorseRank with the allowance checks, is an intended-construct check: it shows coherence within a construct and is not evidence of independent predictive power. A third family, the Sybil-adjusted activity summaries of Section 3.9.3, enters the secondary part of rule A, but no hypothesis rests on it.

The temporal holdout of Section 3.7 provides the out-of-window check. Packin and Lev-Aretz (2024) warn that decentralized credit scores can become opaque; publishing explicit formulas (Section 3.9) and keeping same-window checks apart from the time split reduce that opacity.

3.9 Operational definitions of proxies

In what follows, $\mathcal{W}$ denotes the matched cohort. The notation is that of the check-computation module in the open evaluation pipeline.

3.9.1 Transfer family

Write $\mathcal{T}_{\text{in}}(w)$ for the set of transfers received by contract w in the window and u_e for the sender of transfer e. Self-transfers belong to $\mathcal{T}_{\text{in}}(w)$, so a contract that sends tokens to itself counts itself as one sender, and amounts x_e are summed in US dollars across the five tokens. Define

$$\text{in_degree}(w) = \big|\{\, u_e : e \in \mathcal{T}_{\text{in}}(w) \,\}\big|, \tag{3.10}$$

$$\text{in_value}(w) = \sum_{e \in \mathcal{T}_{\text{in}}(w)} x_e. \tag{3.11}$$

Both are baselines defined in the AWP paper (T. Do et al., 2023), on the premise that economically important addresses receive transfers from more distinct counterparties and in greater total value. Unlike PageRank they are local and ignore the senders' reputations, so comparing AWP with them shows how far propagation, recency weighting and restarts at active senders move its ordering away from the raw inflow.

3.9.2 Allowance family

On the allowance side, w is the spender (owner $\to$ w). With $a^\star$ the latest allowances of Equation (3.7), define

$$\text{in_approve_degree}(w) = \big|\{\, u : a^\star(u, w, \cdot) > 0 \,\}\big|, \tag{3.12}$$

$$\text{in_approve_value}(w) = \sum_{u,\text{token}} \min\big(a^\star(u, w, \text{token}), U\big). \tag{3.13}$$

Like the transfer baselines, they count and sum incoming approvals without following multi-hop endorsement chains; the in-approve value uses the cap U of C-PR.

3.9.3 Sybil-adjusted stability family

The inbound counters leave out self-transfers ($u = w$). Let $N_{\text{tx}}(w)$ be the number of transfers that w receives from other addresses and $N_{\text{uniq}}(w)$ the number of distinct inbound counterparties. Tenure and active months use a second set of rows. Every transfer that involves a cohort contract is attributed to its sender if the sender is in the cohort, and otherwise to its recipient; a transfer between two cohort contracts thus counts only for the sender. We take $t_{\text{min}}(w)$ and $t_{\text{max}}(w)$ to be the earliest and latest timestamps among the rows attributed to w. $N_{\text{mon}}(w)$ is the larger of two counts of distinct calendar months: the months in which w received a transfer from another address, and the months of the rows attributed to w. Define

$$\text{inbound_counterparty_ratio}(w) = \frac{N_{\text{uniq}}(w)}{\max(N_{\text{tx}}(w), 1)}, \tag{3.14}$$

$$\text{transfer_tenure_days}(w) = \frac{t_{\text{max}}(w) - t_{\text{min}}(w)}{86400} \quad \text{(seconds to days)}, \tag{3.15}$$

$$\text{active_months}(w) = N_{\text{mon}}(w). \tag{3.16}$$

The counterparty ratio penalizes contracts that receive many transfers from few addresses, a simple wash pattern. It does not penalize the usual Sybil pattern, in which many fresh addresses send one transfer each; a contract fed that way scores close to the maximum of one. Tenure and active months track how persistent on-chain activity is over the score window. These proxies are therefore weak Sybil checks, and colluding adversaries can satisfy all three.

3.10 Statistical tests

Neither the reputation scores nor the proxies are binary default indicators; both are continuous and are handled as ranks. Following the AWP evaluation protocol (T. Do et al., 2023) and the

approach to construct validity of Cronbach and Meehl (1955), concordance is measured with rank correlations, which are invariant under any monotonic re-scaling of either variable.

- Spearman's ρ: the Pearson correlation of the mid-ranks R_i and S_i of the raw score and the raw proxy value of contract i, tied values sharing the average of their positions (Section 2.9). It measures monotonic agreement: if higher reputation consistently goes with higher proxy values, ρ is positive whatever the nonlinearity in the raw scores.
- Kendall's τ: every reported value is the tie-corrected τ_b of Equation (2.6), computed on the raw scores and proxies. Without ties, $(1+\tau)/2$ is the probability that a randomly chosen pair is ordered the same way by both rankings. With ties that reading belongs to $\tau_a = (n_c - n_d)/n_0$ and not to τ_b. Ties are rare among the PageRank scores of the matched cohort (Section 4.4) but common in the raw degrees and in the labels. When the two sides of a paired contrast tie very different shares of pairs, as a raw count and a PageRank do, τ_b gives the more tied side the smaller denominator, so the contrast can favour it under τ_b and not under τ_a; the degree contrasts and P, computed with τ_b alone, are exposed to this.
- Why both: the AWP protocol (T. Do et al., 2023) reports both, and reporting both keeps the conclusions from hinging on one coefficient. The score distributions are heavy-tailed, with most contracts near zero and a few hubs holding much of the mass, which is why rank coefficients are used instead of a Pearson correlation on raw scores, which the hubs would dominate.
- Family aggregation: A family is summarized by the plain average over its proxies. The transfer-family mean τ, for example, is $\frac{1}{2}(\tau_{\text{in_degree}} + \tau_{\text{in_value}})$. Conclusions are drawn from the confidence intervals, and no absolute τ cutoff is used as a decision criterion.
- Uncertainty: Sampling uncertainty is estimated with a paired bootstrap over contracts (Efron, 1979). The evaluated cohort is resampled with replacement $B = 400$ times (seed 42, fixed in the configuration), and one index matrix is reused for every method and proxy, so the resampled statistics are paired across methods. For each proxy τ, family mean and inter-

method difference, the 2.5th and 97.5th percentiles of the bootstrap distribution form a 95% percentile interval (DiCiccio & Efron, 1996). Because τ_b is a smooth function of U-statistics, the percentile interval is first-order accurate at the sizes of the matched and spender cohorts if contracts are independent; contracts that share owners or belong to one protocol are not, so the intervals may be too narrow. The small number of resamples keeps the $O(B\,n \log n)$ cost of the tie-corrected τ tractable across all methods and proxies, at the price of a Monte Carlo error in each interval end of about 0.13 times the bootstrap standard error. Only the analysis of Section 3.7.3 departs from this, with $B = 2{,}000$.

- Difference tests: Every comparison is a paired contrast on the same resamples. Resample by resample we compute $\Delta\tau = \tau_A - \tau_B$ and report the 95% percentile interval of its bootstrap distribution; a contrast counts as a difference when its interval excludes zero. The intervals of the more than a hundred contrasts computed after the labels were known are not adjusted for multiplicity, and a few would exclude zero by chance alone; a post hoc contrast resolved in one window only is therefore read as a hypothesis, and one resolved in the same direction in both windows as less exposed to chance. Because resampling is paired, the contrast keeps the covariance of the two estimates, $\mathrm{Var}(\hat\tau_A - \hat\tau_B) = \mathrm{Var}\,\hat\tau_A + \mathrm{Var}\,\hat\tau_B - 2\,\mathrm{Cov}(\hat\tau_A, \hat\tau_B)$, and with positive covariance its interval is narrower than the two separate intervals suggest. The contrasts that decide a rule were fixed before the study extracted the data they are judged on: those of rule A (Section 3.7) before any log, and those of rule B and Section 3.7.3 before the W1 logs; the revised analysis applies them after the registered results were known. C-PR against its transfer layer walked alone on new approval pairs in W0 was not among them; it is marked as exploratory wherever it appears. In the same window, eight contrasts set C-PR and S-PR against whichever layer, walked alone, leads each family (the transfer walk on transfer, the allowance walk on allowance, both on Sybil stability), and six set C-PR against AWP on transfer and allowance, against EndorseRank on allowance and transfer and against both on Sybil stability; none of these fourteen was fixed in the plan, and two of them, C-PR against AWP on allowance and against EndorseRank on transfer, were computed afterwards from the

saved scores. Out of window, the contrasts of C-PR with every other score and with the two raw degrees (Table 4.18) were also computed after the labels were known. Five same-window contrasts that involve EndorseRank and AWP, none fixed in advance either, are reported in Appendix 8.5.

- Directionality and imputation: Each proxy is oriented so that a higher value means a better reputation. After re-alignment to the evaluated cohort, a contract absent from a given proxy source is imputed as zero. With sparse transactions this is a conservative choice, and both scoring approaches apply it in the same way. After this step every evaluated contract has a value in every score and proxy vector, so each correlation uses the whole cohort.
- Non-inferiority: Two of the three conditions of rule B ask only that C-PR not fall behind its transfer layer walked alone by more than a margin fixed in advance, $\delta = 0.01$. Such a contrast passes when the lower end of its 95% interval lies above $-\delta$. The test is weaker than a difference test and says nothing about superiority on that label. It is used where the claim under test is that C-PR gives up nothing that matters.
- What is excluded: The study attempts no causal inference, trains no supervised models on separate training and testing sets, and reports no p-values; it asks whether the interval of a contrast excludes the null value.

3.11 Computational benchmark (Claim C5)

Claim C5 concerns the computational cost of each score. The benchmark times the solver call that turns an edge list into scores. It leaves out the one-time cost of building the edge lists, mostly loading and preprocessing, which a scoring service that refreshes its scores periodically would amortize.

- Timed component: The edge lists and AWP's restart weights are built before the timer starts. The timed call indexes the nodes, assembles the sparse weighted adjacency matrix, row-normalizes it, forms the restart vector and runs power iteration with $d = 0.85$, $\varepsilon = 10^{-8}$

and $I_{\max} = 300$. Matrix assembly and each iteration take time that grows with $|E|$ and $|V|$, and a deployment would repeat both at every update. The same graphs give the alignment scores.

- Measurement protocol: Every configuration is measured in one session. An untimed warm-up run, on which alone peak memory is tracked, is followed by five timed runs, from which we report the mean and standard deviation of the run times and the iteration count, with the nodes and edges of the input graph. The same graph under the same settings occurs in the main benchmark, in the $d = 0.85$ row of the damping sweep and in the scaling stage with $n = 6{,}483$, which is the full cohort graph (Section 3.12.1); those tables reuse the one measurement by design, and they are not independent measurements of one quantity. Every runtime table lists edge counts, since edges drive the cost of an algorithm with $O(|E|)$ work per iteration (Langville & Meyer, 2006), and the runtimes are written to the machine-readable output (Section 3.13).
- Cohorts: The main benchmark runs on the graphs of the matched contract cohort ($n = 6{,}483$). For the growth of runtime with graph size we draw deterministic subsamples of the cohort (Section 3.12).
- Fairness: Every algorithm runs with the same solver, tolerance, iteration cap, damping (outside the damping sweep) and seed set, so differences in cost come from the graphs and from AWP's restart vector, which can change the number of iterations, and not from the implementation. Hardware and software are recorded in the benchmark summary, and the cost of two scores is compared through time ratios, each reported beside the corresponding edge ratio.

3.12 Scaling and robustness extensions

We also examine how runtime grows with the size of the graph, and whether the family-level pattern survives changes to the damping, the slope of the value weight and the sample. These analyses qualify Claim C5 and test the reach of the family-level conclusions; neither supplies an alternative

primary sample. Unless stated otherwise, alignment claims in Chapter 4 refer to the matched cohort.

3.12.1 Cohort subsamples for runtime scaling

The scaling stages are drawn from the matched cohort by deterministic SHA256 subsampling with seed string `contract-benchmark-v1`: contracts are sorted by

$$\texttt{SHA256}(\textit{seed}:\textit{address}) \text{ with } \textit{seed} = \texttt{contract-benchmark-v1},$$

that is, by the hash of one string made of the seed, a colon and the lower-cased address, and the subsample is the first n contracts in that order. The seed string is a constant fixed in our configuration. We report four stages, $n = 1{,}000$, $2{,}000$ and $5{,}000$ and the full cohort, $n = 6{,}483$. Since the hashing is deterministic, the subsamples cannot drift between library versions the way pseudorandom draws do, and each stage contains the one before it.

Two features of this design limit what the experiment can show. At every stage the graph keeps the edges of the cohort graph at T_{obs} that have at least one end point among the sampled contracts, and the solver's node set consists of the end points of those edges. The edge count $|E|$ thus grows with the stage but not in proportion to n, since a contract with many counterparties brings many edges. The other is that no stage goes beyond the full cohort graph, so the experiment shows how runtime grows with $|E|$ up to that size and says nothing about larger graphs. Every scaling table reports per-stage $|V|$ and $|E|$ for both methods, and the scaling statement in Chapter 4 goes no further than those columns allow.

3.12.2 Damping sensitivity

On the matched cohort we vary the damping over $d \in \{0.75, 0.85, 0.95\}$, with the graphs and proxies held fixed. A smaller d raises teleportation; a larger one gives multi-hop paths more weight and may slow convergence. The sweep asks whether alignment on the allowance and transfer families depends on the teleportation probability.

3.12.3 Value-slope sensitivity

For this check we recompute EndorseRank and AWP at T_{obs} with the slope b of V multiplied and divided by ten, which moves the amount of half weight to a tenth and to ten times the median. It replaces the plan's twenty-token check, which loses its point with five tokens (Section 3.5.1). The check asks whether the family-level results change when the amount weight rises more or less steeply.

3.12.4 Sample-definition sensitivity

At each scaling stage of Section 3.12.1 the family means of EndorseRank and AWP are recomputed on the stage's contracts, scored on the stage's graph. The stages differ from the cohort in size and in their graphs: a contract's score changes because the edges of the contracts left out of the stage are missing. The per-stage coefficients are therefore a sensitivity check on the sample definition and not evidence of stability, and no intervals were computed for them; the question is how far each coefficient moves when the sample changes. This analysis was not in the plan.

3.13 Reproducibility

No table in Chapter 4 was transcribed by hand. Once the Phase 1 tables are on disk, the evaluation drivers recompute every check, every alignment statistic with its interval, every holdout result and the benchmark, and write them to machine-readable summaries, from which a separate stage exports the tables of Chapter 4 and Appendix 8.1. One versioned configuration file holds the windows, cohort threshold, event-topic IDs, damping, decay parameters, values of λ, bootstrap count and seed, benchmark seed and cost budget, and the revised analysis adds the token list, daily prices and constants of Section 3.5.1 and the start of each score window.

The plan, the registration of W1 and the documents of the revised analysis were committed to the authors' Git repository, which is private. Its folder for this study is published with every commit that changed it in the public repository https://github.com/abitocodes/ERC20-Allowance-Pag

eRank-Wallet-Reputation. Each copied commit keeps the original author, times, message and file contents but has a new identifier, and the file `HISTORY.md` of that repository lists the original identifiers, which the extraction manifest and the results digests record. There the plan is commit `d060f3a` and the registration `1747e64`, and the revised analysis is defined in `experiments/docs/revision_price_weighting.md` and `experiments/docs/revision_one_year_window.md`. The reproducible elements are these: deterministic subsampling of the cohort by SHA256 hashes; a fixed PageRank tolerance and iteration cap; one bootstrap seed and one resample-index matrix for each cohort; the proxy formulas of Section 3.9; the benchmarking procedure of Section 3.11, with the time of every run kept in the output; the coupling weight λ and the holdout logic of Section 3.7, both fixed in the configuration before the study extracted any log; the account types, read at a pinned block; and the extraction manifest of scan B. The machine-readable summaries behind all the tables, with the configuration, the plan and the registration, are committed to that repository under `experiments/data/3-published-results-for-thesis/` (the revised analysis in `revised-usd-12m/`), in the same folder as the source code. Appendix 8.1 lists the GitHub repository URL, the exact commit hash that generated the tables in this thesis, the configuration file and the summary files.

3.14 Sampling considerations and missingness

The matched cohort is a constructed sample, not a random draw from all Arbitrum addresses: eligibility needs non-zero approvals of one of the five tokens from at least three distinct owners in the twelve months before the freeze, and transfer activity plays no part in it. Same-window findings therefore apply to contracts on Arbitrum One that at least three owners approved on the five tokens in these months, and holdout findings to the contract spenders in the cohort's approval data. EOAs and contracts that fewer than three owners approved on these tokens are excluded by design (Section 3.2).

Data can be missing in a few ways. An amount that cannot be parsed is read as zero, which drops a

transfer and resets an allowance. EndorseRank does not see approvals that emit no standard ERC-20 event, and AWP does not see transfers made off-chain or inside custodial accounts. Both are thus biased toward on-chain, standard-token behavior, which is acceptable for an on-chain reputation study but matters once results are extrapolated to all contracts of a protocol.

3.15 Implementation notes

The PageRank code is written in Python, with a SciPy sparse transition operator and NumPy power iteration. Appendix 8.1.1 names the machine on which the benchmark ran. Randomness enters only through the bootstrap, whose seed is fixed; given the downloaded tables and the configuration, evaluation is deterministic. Small floating-point differences between platforms are to be expected, but they would be negligible against the three-decimal-place precision of the τ metric.

3.16 Ethics

Only publicly available blockchain data are used in this research, and it involves no human subjects. The study was reviewed by the institutional ethics committee (Unisa SoC/CAES). Appendix 8.7 contains the ethics clearance certificate (reference [Unisa SoC/CAES ethics clearance ref: TBD], dated [TBD]).

- Privacy and anonymity: An address on Arbitrum One is a pseudonym and does not by itself reveal the real-world identity behind it (Packin & Lev-Aretz, 2024). Pseudonymity is not the same as anonymity, however, and in theory a determined adversary could tie an address to a person through exchange deposits, online profiles or heuristics. In this work we deliberately make no such links. The scored addresses are contracts, and no result is reported for any EOA. Our data collection is limited to on-chain `Approval`, `Transfer` and GMX V2 position events, and we scraped no off-chain personally identifiable information (such as IP addresses, exchange KYC records, social media handles or deanonymization graphs). We did not cluster addresses into real-world entities, publish leaderboards of addresses likely to be personally

identifiable, or try to join an address's transaction history with other databases to infer identity. The published results are aggregate measures only (correlation coefficients, runtimes, family means) and contain no individualized rankings of named participants.

- Data minimization: Scan A keeps every ERC-20 `Approval` and `Transfer` event from 1 October 2023 to 30 June 2026 in the study's cloud project for the cohort rule; every later query and downloaded table is restricted to the cohort's ego network and to the fields that graph construction and the checks require, namely the relevant addresses, token amounts and timestamps. The extracted tables are kept so that the research pipeline can be audited. They will not be released for the purpose of deanonymizing users.
- Data usage and compliance: Every query ran against public data hosted in the `bigquery-public-data.goog_blockchain_arbitrum_one_us` dataset and complied with the Google Cloud terms of service and rate limits. Dry runs and a study budget prevented unnecessary scans. Account types and token data were read with read-only RPC calls, and daily prices came from the public DefiLlama API. We sent no transactions and accessed no privileged or non-public indexing endpoints, and no smart contract was invoked in any way that might interfere with the normal operation of the network.
- Fairness and transparency: Decentralized credit scores raise concerns about fairness as well as opacity (Packin & Lev-Aretz, 2024). New contracts, rarely used contracts and contracts that receive no public approvals may all be undervalued by graph-based reputation systems. Both EndorseRank and AWP carry structural biases of this kind. In the allowance graph, for instance, an address that has received no incoming spend authority is at a disadvantage relative to addresses that have; transfer graphs, in turn, favor addresses embedded in value-flow networks. Chapter 5 discusses the limitations, among them cold-start effects and the Sybil susceptibility of approval graphs.
- Responsible disclosure and impact: EndorseRank may inform risk assessment in DeFi, but it is not financial advice, credit advice or a consumer reporting product. Lending, trading and

compliance decisions still need human oversight. Chapter 5 acknowledges and discusses the potential for adversarial manipulation, for example Sybil attacks on G_{approve} or wash transfers on G_{transfer}. Any future operational deployment would require additional governance, appeal mechanisms and fairness monitoring, all of which lie beyond the scope of this research.

- Summary of methodological commitments: Before results are interpreted, the design has already fixed the chain, windows, cohort and account-type rules, graph formulas, proxy definitions, bootstrap and timing protocols, and statistical tests; Section 3.5.1 describes the one later change. The primary claims concern all the scores together, on the matched contract cohort and the spender holdouts, with each PageRank set against the raw degree it smooths.

Chapter 4 reports what this protocol yields for Arbitrum One.